Lecture notes in Microeconomic Theory: Mechanism and Information Design

1. Mechanism Design

1.1. Why Mechanism design?

1.1.1 Motivating Example: Selling an Indivisible Good

1.1.2 The Economic Environment and Model Setup

1.1.3 Agents, Types, Preferences, and Feasible Outcomes

1.2. Mechanisms and Incentive Compatabilities

1.2.1. Direct and Indirect Mechanisms

1.2.2. Dominant-Strategy Incentive Compatibility

1.2.3. Bayesian Incentive Compatibility

1.2.4. Individual Rationality

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1.4.3. The Vickrey–Clarke–Groves (VCG) Mechanism

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1.5. Further Questions and Research Directions

2. Information Design

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2.1.1. Information as an Economic Instrument

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2.1.4. States of the World, Priors, and Beliefs

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2.2.4. Optimal Information Disclosure

2.3. Bayes Correlated Equilibrium (BCE)

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2.3.3. The BCE Feasibility Problem

2.3.4. Connections to Bayesian Games

2.4. Further Questions and Research Directions

3. The Intersection of Mechanism and Information Design

3.1. Information Asymmetry and Strategic Interaction

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3.3. Mechanism Design with Endogenous Information

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4. Conclusion

4.1. Main Results and Economic Implications

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Preface

These lecture notes are the result of my independent study of microeconomic theory, with a focus on mechanism design and information design. They bring together results from the published literature, original proofs developed as part of my study, and questions for further research.

Theorems, propositions, and lemmas drawn from the literature are attributed to their original sources. Proofs developed independently are explicitly identified as such. Each part concludes with questions intended to encourage further exploration of the topics covered.

Any errors or omissions are my own.

1.1 Motivating example: Selling an indivisible good

Suppose you own an indivisible good, such as a piece of craft, piece of land or a painting. You want to sell it and earn revenue. Your goal is to design a selling mechanism that maximizes your profit. But there is a problem: you do not know the buyer's true valuation v. The buyer knows how much the good is worth to them, but they may have an incentive to misreport their valuation to obtain a lower price.

This raises some important questions:

1. How should you set the price?
2. How can you encourage truthful reporting? If buyers report their valuations, how can you ensure they have no incentive to lie?
3. If several buyers compete for it, who should receive the good?
4. Can you maximize revenue while ensuring the buyer has incentives to participate?

The central idea of mechanism design: rather than simply accepting the market (or game in general) as given, you design the rules of the interaction to achieve a desired outcome despite participants having private information and their own incentives. The challenge is to design rules so that the seller's objectives are achieved while accounting for the buyer's incentives.